

Harshana Lakshara Fernando

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Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering

2022 – 2026

University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream)

2019

St. John's College, Panadura, Sri Lanka

Results: 3 A's (Combined Mathematics, Physics, Chemistry)

District Rank: 17

Experience

R&D

Software Engineer — KōreroCare Platform

Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland | Remote / Part-time

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Software Engineer Intern — Information Systems Associates (Pvt) Ltd

Dec 2024 – May 2025

Aviation Technology | Colombo, Sri Lanka

- **AeroOrder — Distributed PNR Generation Service**
 - Owned design and delivery of a distributed **Spring Boot/Kafka**-backed PNR generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
 - Resolved production **Kafka consumer lag** and offset reset issues using shutdown hooks, retry handling, validation, and a Producer-Consumer refactor.
 - Led the legacy-to-microservice PNR rollout using database-driven **feature toggles** and a **strangler-pattern** migration approach.
 - Built load-testing and monitoring support using **JMeter**, **Prometheus**, and **Grafana** to validate async order/PNR flows, queue time, and service latency.
 - Maintained **>90% unit test coverage** while coordinating delivery with QA and distributed engineering teams.
- **AeroMart — Legacy Reservation System Integration**
 - Integrated **gRPC** clients and **Kafka** consumers into the **Java 8** AeroMart monolith for backward-compatible communication with AeroOrder services.
- **Engineering Practices**
 - Authored comprehensive design docs, sequence diagrams, and integration plans; facilitated knowledge transfer sessions with distributed engineering teams.

- Added **Jenkins** environment support and streamlined service logging by replacing heavyweight logging dependencies with a lightweight structured JSON logging approach.

Projects

See more projects on my portfolio [↗](#)

1. Employee Management System (EMS) (2025 | Individual)

Architecture | Github 

Designed and built a comprehensive, cloud-native microservices application for employee and department management, featuring a full-stack architecture with a React/TypeScript frontend and Spring Boot backend.

- Architected a distributed system using **Spring Cloud** for service discovery, centralized configuration, and an API gateway, with **Resilience4j** for robust fault tolerance.
- Enabled dynamic configuration management using **Spring Cloud Bus** with **RabbitMQ** and established a robust observability stack (Zipkin, ELK, Prometheus/Grafana).
- Designed and managed a CI/CD pipeline using **GitHub Actions**, with **ArgoCD** for GitOps-based deployments to the Kubernetes cluster.

Keywords: Java 17, Spring Boot, Spring Cloud, React, TypeScript, Resilience4j, RabbitMQ, Docker, Kubernetes, GitHub Actions, ArgoCD, Prometheus, Grafana

2. GoTogether (2024 | Group)

Architecture | Live  | Github 

Developed and deployed a full-stack travel assistant platform with a **Next.js** frontend (SSR/CSR, secure auth) and microservices in **Spring Boot** and **Go**, enabling AI itinerary generation via **Gemini API**.

- Implemented a **Spring Boot** social service with **Keycloak** integration for identity management and a dedicated **Go** service using **gRPC** for high-performance inter-service communication.
- Configured production-grade security with **NGINX**, **Certbot**, and **DuckDNS**, and orchestrated deployment on **Oracle Cloud Free Tier** using custom scripts and Docker optimizations.
- Built a dedicated **Auth Service** using **Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Gemini API, Oracle Cloud, gRPC

3. Serverless Auction Platform (2025 | Individual)

Architecture | Github 

A capstone project built using AWS, following a microservices architecture with an event-driven, serverless (FaaS) design.

- Architected an event-driven system using **AWS Lambda**, **API Gateway**, and **EventBridge**, with **SQS** and **SNS** for asynchronous communication.
- Designed a high-performance **DynamoDB** data model with GSIs and implemented a custom **Lambda Authorizer** for JWT-based security.
- Utilized **LocalStack** and **Docker** for local cloud emulation and integrated **Middy** middleware for robust error handling.

Keywords: Node.js, Serverless Framework, AWS Lambda, DynamoDB, SQS, SNS, SES, JWT, Middy, LocalStack, Docker

Other Relevant Projects

4. Automated AI Shorts Video Creator (2025 | Individual)

An open-source project to automate the creation of short-form videos using a powerful combination of AI tools and workflow automation.

- Orchestrated a video creation pipeline using **n8n** as the central workflow engine.
- Integrated various AI services for script generation, text-to-speech, and content preparation, including **Together AI**, **OpenRouter**, and **ElevenLabs**.
- Containerized the entire stack using **Docker Compose** for one-click local deployment.

Keywords: n8n, Docker, AI, Workflow Automation, Video Generation

5. Food Delivery Application (2024 | Group)

Github 

Built a scalable, event-driven food delivery platform inspired by systems like Uber Eats, using a modern microservices architecture. As part of the core team, I focused on designing and developing the Restaurant Service.

- Developed the Restaurant Module with secure sign-up/login using **Spring Security** and **JWT** (with extended claims).
- Built CRUD APIs for menu and item management; integrated **Redis caching** to improve performance and reduce database load, with appropriate cache invalidation strategies.
- Enabled order sync via **Kafka**: when an order is placed, an event is published to the 'order_created' topic, which the restaurant service consumes to update order details.

Keywords: Java, Spring Boot, gRPC, Kafka, Redis, PostgreSQL, WebSocket, Docker, React.js, Ant Design

6. Go gRPC GraphQL – Multi-Tenant Microservices (2025 | Individual)

Architected a **polyglot, multi-tenant microservices** platform with a focus on scalability, fault isolation, and unified data access.

- Developed backend services in **Go** using **gRPC** and **Protobuf** for efficient inter-service communication.
- Implemented a **GraphQL API Gateway** to provide a single, unified endpoint for client applications.
- Designed multi-tenant data storage with **PostgreSQL**, **Elasticsearch**, and optional **Cassandra** for high-volume workloads.

Keywords: Go, gRPC, GraphQL, Protobuf, PostgreSQL, Elasticsearch, Cassandra, Docker Compose

7. RPAL Interpreter – Functional Language Compiler (2024 | Group)

Engineered a complete compiler and interpreter for the RPAL (Right-reference Pedagogic Algorithmic Language), built from scratch to parse and evaluate programs according to official language grammar and semantics.

- Developed a Lexical Analyzer using custom Finite State Automata (FSA) and regex-based token identification.
- Built an Abstract Syntax Tree (AST) and Standard Tree (ST) generator.
- Implemented a CSE Machine to evaluate programs using 13 semantic rules of RPAL.
- Built a robust testing framework with **Pytest** covering over 168 test cases across AST, ST, and full output validation.

Keywords: Python, Pytest, GitHub Actions, Makefile, CLI

Professional Training, Coursework & Certificates

See all certificates

- **Certified Kubernetes Administrator (CKA) Training**
CodeKloud (Mumshad Mannambeth) — Completed 27h curriculum with 50+ hands-on labs.
 - **AWS Certified Solutions Architect Associate (SAA-C03) Training**
Udemy (Stephane Maarek) — Completed 27.5h comprehensive cloud architecture course.
 - **Apache Kafka Series (v3) – Learn Kafka for Beginners**
Udemy (Stephane Maarek) — Full System Design & Setup.
 - **100 Days of DevOps (Bootcamp)**
CodeKloud — Intensive hands-on training covering Linux, Python, Docker, Jenkins, Terraform & Ansible.
 - **Implementing Software Architecture for Large-Scale Systems**
Udemy
 - **Advanced OAuth Security**
Udemy
 - **Next.js: Build Scalable React Apps with Page & App Routers**
Udemy — Instructors: Anton Voroniuk, Denys Kohut, Anton Voroniuk Support
- Apr 2025

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- Aug 2025

View certificate
- Mar 2025

[Part 1] | [Part 2] | [Part 3]
- Nov 2025

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- Apr 2025

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- Mar 2024

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- Nov 2025

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Skills

Core	Java Spring Boot Microservices Distributed Systems System Design REST APIs
Backend	Spring Cloud Spring Security Resilience4j Node.js Express.js Gin
Messaging & APIs	Kafka RabbitMQ gRPC GraphQL WebSockets Protobuf JWT OAuth 2.0
Data	PostgreSQL MySQL Oracle DB Redis Couchbase
Cloud & DevOps	AWS Docker Kubernetes CI/CD (GitHub Actions, Jenkins)
Testing	JMeter Gatling Pytest Unit Testing
Tools	Git Gradle Postman Jira Confluence

References available upon request.